Pathfinders

Pathwise AI
Project Vision Document

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15/02/2026

Release Checklist

Checklist Item	Control (Yes/No)
The document has passed the quality assurance process.	YES
The document is visually appealing and has a professional look and structure.	YES
All placeholders such as text within <> are replaced with appropriate text.	YES
The document is prepared in the “Justify Text” format.	YES
The document does not include typos, different fonts, and misalignments.	YES
In the revision history, the first version states “The first version”. Additional versions include an entry highlighting all the changes from the previous version.	YES
The table of contents (TOC) is updated as necessary.	YES
References list other project documents and additional standards or documents needed to understand the document. The references include the necessary document/artifact version number and publication date.	YES
The language grammar is used appropriately.	YES
Sentences are written as full sentences.	YES
The wording in the document is formal.	YES
The wording in the document is clear.	YES
The wording in the document is professional.	YES
All project-related definitions, acronyms, and abbreviations sections are included in the related tables.	YES
The team member names are specifically spelled out in the document preparation and approvals sections.	YES
The purpose and scope of the document are adequately described.	YES
All sections include the necessary and adequate information.	YES
All stakeholders and users are identified as related to the product.	YES
The user environment is adequately analyzed and described.	YES
The product alternatives and competitive products are adequately analyzed.	YES

The "Precedence and Priority" section includes the list of features in a priority order.	YES

Final Control	**Yes/No**
The document is checked based on the "Team Project Artifacts Common Errors" Guideline. The document is ready for release.	YES
The document is not ready for release.	NO

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Project Vision Document

1. Introduction

1.1 Purpose of the Document

The purpose of this document is to define and communicate the high-level vision, needs, and features of the PathWise AI system. The capabilities required by stakeholders and target users are identified, along with the rationale for these needs. This document establishes a shared understanding of the product direction and scope. Detailed functional behavior and implementation details are intended to be specified in subsequent requirements and design documents.

1.2 Scope of the Document

This document provides a comprehensive overview of the PathWise AI system, including its business opportunity, problem statement, stakeholder descriptions, product overview, and detailed feature specifications. It serves as the foundational vision document that guides all subsequent development activities, including requirements gathering, design, implementation, and testing phases. The document is intended for all project stakeholders including the development team, project sponsors, and potential end users.

1.3 References

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1.4 Definitions, Acronyms, and Abbreviations

Term	Abbreviation / Acronym	Definition
Software-Defined Wide Area Network	SD-WAN	A virtual WAN architecture that allows enterprises to leverage any combination of transport services to securely connect users to applications.
Software-Defined Networking	SDN	An approach to networking that uses software-based controllers or APIs to direct traffic on the network and communicate with the underlying hardware.
Long Short-Term Memory	LSTM	A type of recurrent neural network architecture designed to learn long-term dependencies in sequential data.
Intent-Based Networking	IBN	A form of network administration that incorporates machine learning and orchestration to automate administrative tasks.
Voice over Internet Protocol	VoIP	A technology that allows voice calls to be made over the internet.
Small-to-Medium Enterprise	SME	Organizations with fewer than 500 employees and revenue typically under $50 million.
Application Programming Interface	API	A set of protocols and tools for building software applications.
Command Line Interface	CLI	A text-based interface used to interact with software and operating systems.
Digital Twin	DT	A virtual representation of a physical system that can be used for simulation and testing.

2. Positioning

2.1 Business Opportunity

The enterprise networking market faces a critical challenge: traditional SD-WAN solutions react to network problems after they occur, causing dropped connections, degraded performance, and business disruption. PathWise AI addresses this gap by providing predictive network intelligence that anticipates issues before they impact users. The target market of Small-to-Medium Enterprises (SMEs) represents a significant opportunity, as these organizations require enterprise-grade reliability without the prohibitive costs of proprietary hardware and specialized networking staff. By offering a vendor-agnostic, AI-powered solution with an intuitive natural language interface, PathWise AI removes traditional barriers to advanced network management, making sophisticated SD-WAN capabilities accessible to organizations with generalist IT teams.

2.2 Problem Statement

The problem of	the "switching gap" in traditional networking, where network performance degrades or connections drop before backup links are engaged, resulting in service disruption for mission-critical applications.
affects	network administrators, IT teams, employees using real-time applications (VoIP, video conferencing), healthcare facilities requiring uninterrupted connectivity, enterprises dependent on cloud platforms, and retail chains with distributed locations
the impact of which is	dropped calls during critical meetings, interrupted surgical video feeds, failed financial transactions, lost productivity, customer dissatisfaction, and potential revenue loss due to network downtime.
a successful solution would be	an intelligent system that predicts network degradation 30-60 seconds before it occurs, automatically reroutes traffic without packet loss (hitless handoff), validates all changes in a safe sandbox environment before deployment, and provides an intuitive interface that does not require specialized networking certifications.

2.3 Product Position Statement

For	Small-to-Medium Enterprises (SMEs), Managed Service Providers, Healthcare Facilities, Educational Institutions, and Retail Chains requiring multi-site network connectivity.
Who	need reliable, uninterrupted network performance for mission-critical applications without the cost and complexity of traditional enterprise solutions.
The Pathwise AI	is an AI-powered SD-WAN management platform.
That	predicts network degradation before it occurs and automatically optimizes traffic routing to prevent disruptions, while validating all changes in a digital twin environment for operational safety.
Unlike	traditional SD-WAN solutions (Cisco Viptela, VMware VeloCloud, Fortinet) that react to network issues after they occur and require expensive proprietary hardware and certified specialists.
Our product	uses LSTM deep learning for predictive analytics, offers vendor-agnostic compatibility, provides natural language policy management (Intent-Based Networking), and includes built-in safety validation through Digital Twin simulation.

3. Stakeholder and User Descriptions

To effectively provide products and services that meet stakeholders' and users' real needs, it is necessary to identify and involve all stakeholders as part of the requirements modeling process. This section provides a profile of the stakeholders and users involved in the project, and the key problems they perceive to be addressed by PathWise AI.

3.1 Stakeholder Summary

Name	Description	Responsibilities
Course Instructor	The professor teaches COSC6370-001 (Advanced Software Engineering) who oversees and evaluates the project.	Evaluates project deliverables and milestones Ensures the project follows sound software engineering practices Approves project scope and requirements Monitors the team's progress throughout the semester Provides academic guidance and feedback
SME Business Owners / IT Decision Makers	Owners and IT leaders of Small-to-Medium Enterprises who would fund and adopt PathWise AI for their organizations.	Approves funding and procurement of the SD-WAN solution Ensures the system meets business continuity requirements Evaluates ROI and cost-effectiveness vs. proprietary solutions Defines high-level network reliability expectations
Network Administrators & IT Teams	Technical professionals responsible for day-to-day network monitoring, configuration, and troubleshooting.	Monitors network health using the Multi-Link Health Scoreboard Configures traffic steering policies via the IBN interface Reviews and approves AI-driven routing changes before deployment Ensures network security and compliance with organizational policies
Managed Service Providers (MSPs)	Third-party providers managing network infrastructure for multiple client organizations.	Deploys and manages PathWise AI across multiple client sites Ensures service-level agreements (SLAs) are met Reduces manual intervention and support tickets through proactive management Monitors predictive alerts to prevent client-facing outages
End Beneficiaries	Employees, healthcare professionals, educators, and retail staff who rely on stable network connectivity.	Utilizes real-time applications (VoIP, video conferencing, cloud platforms) Reports network issues when experienced Benefits from uninterrupted connectivity without direct system interaction

Network Security & Compliance Officers	Professionals responsible for ensuring that AI-driven routing changes comply with security policies and regulations.	Validates that routing changes comply with firewall and security policies Ensures the Digital Twin Sandbox properly tests changes before production deployment Monitors the system for potential security vulnerabilities Ensures regulatory compliance across network operations

3.2 User Summary

Name	Description	Responsibilities	Stakeholder
Network Administrator	Primary technical user who monitors and manages the SD-WAN system on a daily basis.	Monitors real-time network telemetry and link health scores Configures traffic routing policies and priorities Reviews AI-generated predictions and recommended actions	Network Administrators & IT Teams
		Validates routing changes in the Digital Twin Sandbox Responds to predictive alerts and system notifications	
Generalist IT Staff	SME IT personnel without specialized networking certifications who manage the system using simplified interfaces.	Manages network policies using natural language commands via the IBN interface Sets business-level priorities (e.g., “Prioritize VoIP over guest Wi-Fi”) Monitors the Multi-Link Health Scoreboard for network status Escalates complex issues when needed	SME Business Owners / IT Decision Makers
MSP Technician	Service provider staff managing PathWise AI deployments across multiple client networks.	Deploys and configures the system for client environments Monitors predictive alerts across multiple client sites Manages SLA compliance and generates performance reports Coordinates with clients on network policy changes	Managed Service Provider (MSPs)

IT Manager / Business Leader	Decision maker who oversees network strategy and reviews AI transparency reports.	Reviews AI decision-making reports and confidence levels Approves high-level network policies and configurations Monitors overall network performance and uptime metrics Evaluates system ROI and cost savings	SME Business Owners / IT Decision Makers
End User (Employee)	Indirect user who benefits from improved network reliability for daily operations.	Uses real-time applications (VoIP, video, cloud services) Reports connectivity issues to IT when experienced Benefits from seamless network handoffs without direct interaction	End Beneficiaries

3.3 User Environment

The target user environment for PathWise AI encompasses the following characteristics:

- Team Size: Network operations teams typically range from 2-10 members in SME environments, with potential growth as organizations scale.
- Task Cycle: Network monitoring is continuous (24/7), with configuration changes occurring weekly or monthly. Incident response activities may last from minutes to hours depending on severity.
- Environment Constraints: Users operate from central IT offices, remote locations, and mobile devices. The system must support multi-site management with varying connection types (Fiber, Satellite, 5G, Broadband).
- Current Platforms: Organizations use a mix of Windows, Linux, and macOS workstations. Network infrastructure includes various vendor equipment (Cisco, Juniper, Arista, etc.).
- Integration Requirements: PathWise AI must integrate with existing monitoring tools (SNMP, NetFlow), SDN controllers (OpenDaylight, ONOS), and enterprise systems (ticketing, CMDB).

3.4 Summary of Key Stakeholder or User Needs

The following section lists the key problems with existing solutions as perceived by stakeholders and users. For each problem, the reasons, current solutions, desired solutions, and relative importance are described.

3.4.1 Reactive Network Failure Detection Causes Downtime and Dropped Packets

Traditional SD-WAN systems rely on threshold-based monitoring that only triggers failover after degradation has already occurred, creating the “switching gap” where packets are lost during the transition. Currently, network administrators manually monitor dashboards and set static threshold alerts. When a link degrades, failover is triggered reactively, often resulting in dropped VoIP calls, frozen video, and service interruptions. Stakeholders want a predictive system using LSTM-based deep learning that forecasts link brownouts 30–60 seconds in advance and executes “hitless” traffic steering with zero packet loss before degradation impacts users. This is a Critical (Must Solve) need, as it directly addresses the core problem PathWise AI is designed to eliminate.

3.4.2 Complex CLI-Based SD-WAN Management Requires Specialized Expertise

Configuring SD-WAN policies requires knowledge of CLI commands, YANG/NETCONF, and vendor-specific syntax. SMEs often lack staff with specialized networking certifications such as CCNP or JNCIS. Currently, organizations either hire expensive certified network engineers, outsource to managed service providers, or use simplified but limited consumer-grade routers that lack advanced traffic steering capabilities. Stakeholders want an

Intent-Based Networking (IBN) interface that allows generalist IT staff to manage complex network policies using natural language commands, such as "Prioritize medical imaging traffic over guest Wi-Fi." This is a Critical (Must Solve) need, as it removes the primary barrier to SME adoption.

3.4.3 AI-Driven Routing Changes Risk Destabilizing the Production Network

Autonomous AI decisions are perceived as "black box" operations. Without validation, an incorrect routing change could introduce loops, violate firewall policies, or crash the live network. Currently, network changes are tested manually in lab environments or pushed directly to production with rollback plans, with no automated pre-deployment validation for AI-generated routing decisions. Stakeholders want a Digital Twin Validation Sandbox using Mininet and Batfish that simulates every AI-driven routing change in a virtual environment, verifying it is loop-free and policy-compliant before production deployment. This is a Critical (Must Solve) need, as operational safety is essential for enterprise trust and adoption.

3.4.4 Lack of Transparency in AI Decision-Making Reduces Trust

Network administrators and business leaders need to understand why the AI made a specific routing decision. Without explainability, adoption is hindered by a lack of trust in autonomous systems. Most existing SD-WAN solutions provide basic logging but offer no insight into why a particular path was chosen or what predictive confidence levels guided the decision. Stakeholders want a Multi-Link Health Scoreboard providing real-time visualization of link health scores, AI prediction confidence levels, and clear reasoning behind every automated path switch. This is a High (Should Solve) need, as transparency directly influences stakeholder confidence and long-term adoption.

3.4.5 Enterprise-Grade SD-WAN Solutions Are Cost-Prohibitive for SMEs

Proprietary SD-WAN hardware and licensing from vendors like Cisco Viptela or VMware VeloCloud require significant capital expenditure, making them inaccessible for small-to-medium enterprises. Currently, SMEs either invest heavily in proprietary solutions, use basic consumer-grade networking equipment, or operate without intelligent traffic management, accepting frequent downtime as a cost of business. Stakeholders want a vendor-agnostic, software-based SD-WAN solution that delivers enterprise-grade predictive reliability on commodity hardware, significantly reducing total cost of ownership. This is a High (Should Solve) need, as affordability is central to PathWise AI's market positioning.

3.4.6 Managing Multiple WAN Links Across Distributed Locations Is Operationally Complex

Organizations with multiple branches use diverse WAN links such as Fiber, Satellite, and 5G, each with different performance characteristics. Manually managing failover policies across all locations is unsustainable. Currently, MSPs and IT teams use separate monitoring tools for each link type, manually configure failover priorities per site, and rely on reactive ticket-based support when issues arise. Stakeholders want unified multi-link health monitoring with autonomous, predictive traffic steering across all WAN paths managed from a single centralized dashboard. This is a Medium (Nice to Solve) need, as it enhances operational efficiency but is not a core blocker to initial adoption.

3.5 Alternatives and Competition

The following alternatives and competitive solutions exist in the SD-WAN market:

- Cisco Viptela: Enterprise-grade SD-WAN with strong security features. Strengths include broad feature set and enterprise support. Weaknesses include high cost, proprietary hardware requirements, and reactive rather than predictive approach.
- VMware VeloCloud: Cloud-first SD-WAN solution. Strengths include good cloud integration and scalability. Weaknesses include complexity for SMEs and limited AI/ML capabilities for prediction.
- Fortinet Secure SD-WAN: Security-focused SD-WAN. Strengths include integrated security and competitive pricing. Weaknesses include limited predictive capabilities and the Fortinet ecosystem.
- Manual Management (Status Quo): Traditional router/switch management with SNMP monitoring. No additional cost but requires significant expertise, lacks automation, and provides no predictive capability.

4. Product Overview

4.1 Product Perspective

PathWise AI operates as an intelligent overlay management layer for existing network infrastructure. It is not a standalone network device but rather integrates with existing SDN controllers (OpenDaylight, ONOS), network equipment via standard protocols (NETCONF, YANG, SNMP), and telemetry sources to provide predictive intelligence and autonomous optimization. The system consists of three primary components:

(1) the Telemetry Ingestion Layer that collects real-time network metrics,
(2) the AI Prediction Engine powered by LSTM neural networks, and
(3) the Control Interface that executes routing decisions through SDN controllers. The Digital Twin Validation Sandbox operates in parallel to test all proposed changes before production deployment.

4.2 Assumptions and Dependencies

The following assumptions and dependencies apply to PathWise AI:

- Organizations have or will deploy SDN-capable network infrastructure supporting OpenFlow or similar protocols.
- Network devices support telemetry collection via SNMP, NetFlow, or streaming telemetry protocols.
- Sufficient historical network data (minimum 30 days) is available for initial LSTM model training.
- Organizations have multiple WAN links (Fiber, Satellite, 5G, Broadband) to enable traffic steering capabilities.
- Mininet and Batfish frameworks remain available and compatible for Digital Twin simulation.SDN controllers (OpenDaylight or ONOS) maintain API compatibility with current versions.

4.3 Cost and Pricing

PathWise AI is designed as a "Lean IT" solution with cost structures appropriate for SME budgets. The platform utilizes open-source components (OpenDaylight, Mininet, Batfish) to minimize licensing costs. Pricing will be based on a subscription model tied to the number of managed network sites and WAN links. Implementation costs include initial deployment, network integration, and LSTM model training. Ongoing operational costs include cloud hosting for the AI engine, support services, and periodic model retraining. The vendor-agnostic approach eliminates hardware lock-in costs associated with proprietary solutions.

4.4 Licensing and Installation

PathWise AI will be offered as a Software-as-a-Service (SaaS) solution with optional on-premises deployment for organizations with data residency requirements. Installation requires deployment of lightweight telemetry agents on network devices and integration with existing SDN controllers. The IBN interface is web-based and requires no client software installation. Enterprise deployments may require professional services for initial network assessment, integration configuration, and AI model customization.

5. Product Features

5.1 Feature 1 – Predictive Telemetry Engine

The Predictive Telemetry Engine utilizes a Deep Learning-based LSTM (Long Short-Term Memory) neural network to ingest high-frequency network metrics including latency, jitter, and packet loss. Unlike traditional systems that react to current data, this feature forecasts link performance up to 60 seconds into the future, identifying "brownout" patterns before they disrupt the user experience. The engine continuously analyzes telemetry streams from all WAN links and generates predictive health scores that enable proactive traffic management decisions.

5.2 Feature 2 – Autonomous Traffic Steering (Hitless Handoff)

The Autonomous Traffic Steering feature integrates directly with SDN controllers (OpenDaylight or ONOS) to dynamically modify flow tables based on predictive intelligence. This enables the system to execute "hitless" handovers, shifting high-priority packets (VoIP, Video, or Financial transactions) from a degrading link to a stable

alternative with zero packet loss and no manual intervention. The feature maintains session state during transitions to ensure seamless user experience for mission-critical applications.

5.3 Feature 3 – Digital Twin Validation Sandbox

The Digital Twin Validation Sandbox serves as a critical safety layer that mirrors the live network environment using Mininet and Batfish. Every AI-driven routing change is first deployed and tested in this virtual "sandbox" to ensure it is loop-free, compliant with existing firewall policies, and will not introduce network instability before being pushed to the production infrastructure. This feature addresses the enterprise concern about "black box" AI decisions by providing transparent validation and audit trails.

5.4 Feature 4 – Intent-Based Management Interface (IBN)

The Intent-Based Management Interface provides a simplified, natural-language-ready dashboard that translates high-level business goals into complex YANG/NETCONF configurations. For example, an administrator can specify "Prioritize medical imaging traffic over guest Wi-Fi" without understanding the underlying protocol details. This feature removes the "CLI barrier," making advanced SD-WAN management accessible to generalist IT staff without requiring specialized certifications such as CCNA or CCNP.

5.5 Feature 5 - Multi-Link Health Scoreboard

The Multi-Link Health Scoreboard provides a real-time visualization layer that displays a "Health Score" for every available WAN path (Fiber, Satellite, 5G, Broadband). It offers a transparent view of the AI's decision-making process, showing predictive confidence levels and the reasoning behind every automated path switch. This feature enables network administrators to understand and trust the AI's decisions while maintaining oversight of network operations.

6. Constraints

The following constraints apply to the PathWise AI system:

- Technical Constraints: The system requires SDN-capable network infrastructure. Legacy networks without SDN support cannot utilize the autonomous traffic steering feature.
- Data Requirements: LSTM model accuracy depends on historical network data. A minimum of 30 days of telemetry data is required for initial training, with ongoing data collection for model refinement.
- Latency Constraints: Prediction and steering decisions must execute within milliseconds to enable hitless handoffs. This constrains the complexity of AI models that can be deployed in real-time.
- Regulatory Constraints: Healthcare deployments must comply with HIPAA requirements for data handling and audit logging. Financial services may have additional compliance requirements.
- Resource Constraints: The development team has four members with a defined project timeline aligned with the academic semester.
- Integration Constraints: The system must maintain compatibility with multiple SDN controller versions and vendor equipment, requiring extensive testing across configurations.

7. Quality Ranges

Quality Attribute	Target Range	Description
Prediction Accuracy	>=90%	The LSTM model must achieve at least 90% accuracy (measured by Mean Squared Error) for 30-60 second latency/jitter/packet loss predictions.
System Availability	>= 99.9%	PathWise AI management platform must maintain high availability to ensure continuous network optimization.

Failover Time	< 50 ms	Traffic steering decisions must execute within 50 milliseconds to achieve hitless handoff without packet loss.
Sandbox Validation	< 5 seconds	Digital Twin validation must complete within 5 seconds to enable timely production deployment of routing changes.
User Interface Response	< 2 seconds	IBN dashboard operations and natural language query processing must respond within 2 seconds.
Scalability	>= 100 sites	The system must support management of at least 100 network sites with multiple WAN links each.
Usability	No CLI required	All common administrative tasks must be accomplishable through the IBN interface without command-line knowledge.

8. Precedence and Priority

Feature	Priority (High/Medium/ Low)	Priority Reasoning
Predictive Telemetry Engine	High	This feature has the highest priority since it is the core differentiator of PathWise AI. Without accurate predictions, the system cannot proactively prevent network issues. This is the foundation upon which all other features depend.
Autonomous Traffic Steering	High	This feature has high priority since it delivers the primary user benefit: hitless handoffs. The predictive intelligence is only valuable if the system can act on predictions to reroute traffic before degradation occurs.
Digital Twin Validation Sandbox	High	This feature has high priority since it addresses the critical enterprise requirement for operational safety. Without sandbox validation, organizations cannot trust AI-driven changes, limiting adoption in production environments..
Multi-Link Health Scoreboard	Medium	This feature has medium priority since it provides essential visibility for administrators but is not required for core functionality. The system can operate without the dashboard, though user trust and oversight would be reduced.
Intent-Based Management Interface	Low	This feature has low priority since it significantly improves usability for non-specialist administrators. While important for market differentiation and SME adoption, the system can function with traditional configuration methods.